Lecture 8: (Abbreviated) Lasso Regression & Catch-up

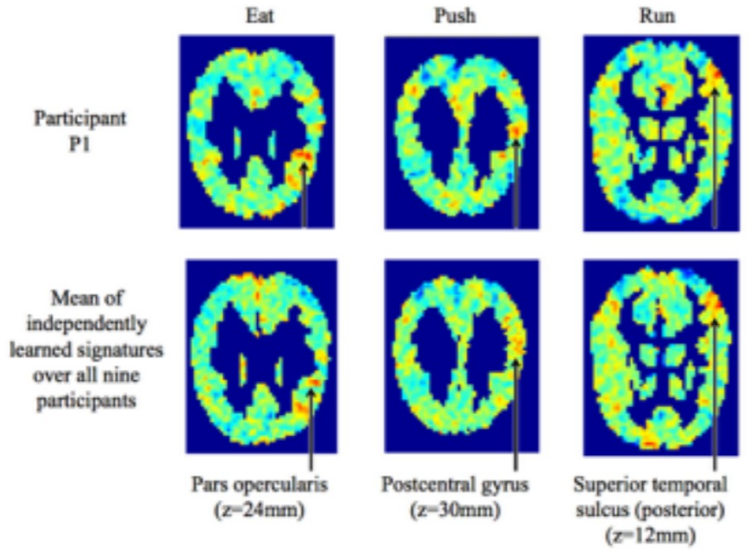
“Sparsity”

- Vector w is sparse if many entries are zero
  - For example, if you roll a single die 3 times and record how many 1’s, 2’s, 3’s, 4’s, 5’s, and 6’s, you will have a sparse vector
  - In reality entries close to zero are sometimes considered “sparse”
    - Two definitions depending on context!!

- Useful in many tasks
  - Efficiency: if size(w) is a billion, each prediction is expensive:
    - If part of an online system, this can be very slow
    - If w is sparse, prediction computation only depends on number of non-zeros

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  - Interpretability: what are the relevant dimensions to make a prediction?
    - What are the parts of the brain associated with particular words?



Question: How do we find the “best” subset among all possible features?

Greedy model selection

- Pick features
- Greedy heuristic:
  - Start from an empty (or simple) set of features, F_0 = null
  - Run learning algorithm for current set of features, F_t
    - Obtain weights for these features
  - Select next best feature, $h_i(x)^*$
    - That is, $h_j(x)$ that results in the lowest training error when using $F_t + \{h_j(x)^*\}$
  - F_{t+1} <- $F_t + \{h_i(x)^*\}$
  - Recurse

Greedy model selection

- Applicable in many situations (some of which we'll talk about later in this course):
  - Logistic regression: selecting features (basis functions)
  - Naïve Bayes: selecting (independent) features $p(X_i|Y)$
  - Decision trees: selecting leaves to expand
- Only a heuristic
  - Finding the best set of k features is computationally intractable!

Greedy model selection

- Recurse until??? When do you stop?
  - When training error is low enough?
  - When test set error is low enough?
  - Using cross validation?

Is there a set solution that makes sense?

Lasso regression

- **L**east **A**bsolute **S**hrinkage and **S**election **O**perator
- Finds the coefficients that should be used (like linear regression)
- Also provides best subset selection, which indicates the most valuable variables for using for machine learning